

PIPE THICKNESS LOSS ANALYSIS

Market Pipe Dataset • 1,000 Records • 11 Variables

487

CRITICAL ASSETS



KEY HEADLINE: Nearly half the pipes are classified **Critical**, and thickness loss dominates the condition signal.

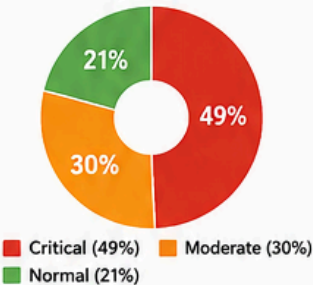


1 DATA QUALITY & PROFILE

The dataset is clean; no missing values or duplicate rows were detected.

Records	1,000
Variables	11
Materials	5
Grades	5
Pipe sizes	13
Missing values	0
Duplicate rows	0

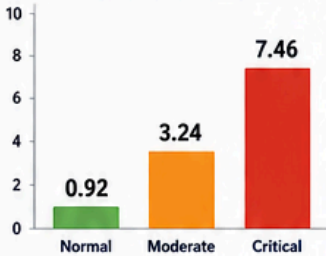
CONDITION SPLIT



2 CONDITION IS STRONGLY SEPARATED BY THICKNESS LOSS

Average thickness loss rises sharply from Normal to Moderate to Critical.

AVERAGE THICKNESS LOSS (mm)



INTERPRETATION

- Critical assets average 7.46 mm of thickness loss — more than 8x the Normal average of 0.92 mm.
- The practical triage threshold is clear: assets above ~5 mm loss are predominantly Critical.
- Priority cue: focus inspection capacity on high thickness-loss assets first.

3 PRIMARY DRIVERS OF CONDITION CLASSIFICATION

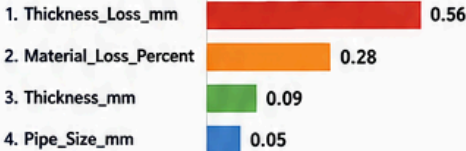
A random-forest classifier confirms that direct loss metrics explain the condition label.

MODEL READOUT



5-fold cross-validated accuracy:
99.8%

TOP DRIVERS (Feature Importance)

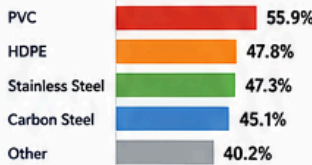


Pressure, temperature, corrosion impact, and age showed weak standalone correlations with thickness loss.

4 MATERIAL AND GRADE RISK CONCENTRATION

PVC has the highest Critical rate; API 5L X65 and X52 rank highest among grades.

BY MATERIAL (% CRITICAL)



BY GRADE (% CRITICAL)

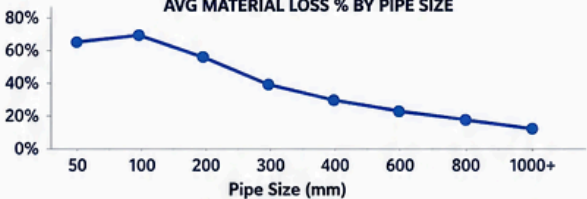


Use these segments for risk-based sampling, but avoid over-weighting material alone: direct loss metrics are much stronger indicators.

5 PIPE SIZE LENS: SMALL PIPES SHOW HIGHER MATERIAL LOSS PERCENTAGE

Percentage loss is inflated for smaller baseline wall thickness, even when mm loss is similar.

AVG MATERIAL LOSS % BY PIPE SIZE



INSIGHT: Average material-loss percentage is highest at 50 mm and 100 mm pipe sizes. Smaller pipes have thinner starting walls; the same millimeter loss translates into a larger percentage loss.



RECOMMENDATION: Use both absolute mm loss and percentage loss in triage rules.

6 RECOMMENDED ACTION PLAN – PRIORITIZED BY RISK AND CONSEQUENCE

Use condition severity, consequence proxies, and deterioration velocity to sequence work.

1

TIER 1: INTERVENE ON HIGH-CONSEQUENCE CRITICAL ASSETS

Inspect the 269 Critical records that also operate at $\geq 1,500$ psi or have diameter ≥ 800 mm. Treat this as the immediate integrity queue.



2

TIER 2: ADDRESS REMAINING CRITICAL AND NEAR-CRITICAL ASSETS

Schedule the remaining 218 Critical records next, then review 108 Moderate records already at ≥ 4.5 mm loss or $\geq 50\%$ material loss.



3

TIER 3: INSTITUTIONALIZE RISK-BASED INTEGRITY MANAGEMENT

Add location, service fluid, coating, inspection dates/method, repair history, shutdown cost, and minimum allowable wall thickness to quantify probability and consequence of failure.



BOTTOM LINE: Operational risk is concentrated in high-loss assets, not broadly explained by environment variables in this dataset.

7 A FOUR-TIER INTERVENTION QUEUE CONCENTRATES EFFORT WHERE IT MATTERS

Severity is combined with pressure and diameter as practical consequence proxies.

IMMEDIATE
Critical + high consequence proxy



269
27%

NEXT
Remaining Critical



218
22%

WATCHLIST
Moderate near Critical



108
11%

ROUTINE
Remaining population



405
40%

BUSINESS INSIGHT

- ✓ 27% of the portfolio sits in the immediate queue, rather than treating all 49% Critical records as equally urgent.
- ✓ This sequencing can reduce exposure faster and makes inspection capacity, shutdown planning, and maintenance budgets easier to defend.

DETERIORATION VELOCITY IDENTIFIES ASSETS THAT MAY OUTFRAN THE INSPECTION CYCLE

Loss per service year is a screening indicator; validate it against actual inspection intervals.



182 assets ≥ 1.0 mm/year
18% of portfolio



144 Critical + fast loss
79% of fast-loss assets



108 Moderate near Critical
Need earlier reinspection

MEDIAN THICKNESS-LOSS RATE (mm/year)



OPERATIONAL RECOMMENDATION

- ✓ Set inspection intervals by observed deterioration velocity, not age alone.
- Fast-loss Critical: engineering review and confirmatory inspection now
- Near-Critical Moderates: shorten reinspection interval
- Stable Normal/Moderate: retain routine cadence

Age is not a strong separator in this dataset; a uniform age-based replacement rule would misallocate capital.

BUSINESS IMPLICATION: Dynamic intervals can prevent surprise escalation while reducing unnecessary inspections on stable assets.

CONVERT THE ANALYSIS INTO AN INTEGRITY-MANAGEMENT OPERATING MODEL

The dataset supports prioritization; additional fields are needed for defensible failure probability and financial impact.



0-30 DAYS Stabilize Exposure

- Validate the 269 immediate assets
- Confirm wall-thickness readings
- Escalate fast-loss Critical cases



30-90 DAYS Optimize the Workbank

- Schedule remaining 218 Critical
- Reinspect 108 near-Critical Moderate
- Bundle work by outage/location



90+ DAYS Build Predictive Control

- Capture minimum allowable thickness
- Add failure/repair and cost history
- Track risk reduction after intervention

TRACK BUSINESS OUTCOMES



High-risk backlog reduced



Inspection cycle time



Emergency repairs avoided



Planned vs. unplanned downtime



Risk-weighted maintenance spend

⚠ Do not use this model as a replacement/repair decision by itself; engineering limits and field validation remain required.



Navigating Human Bias

as an AI Leader in the Energy Industry

Great AI starts with great leadership.
Technology is a tool. People and judgment lead the way.



MY LEADERSHIP VALUES

I believe AI should enhance—not replace—human judgment. My core value is **responsible innovation**: using AI with transparency, fairness, and accountability.

In oil and gas, AI helps us optimize production, predict equipment failures, and manage reservoirs better. But AI systems reflect the decisions, assumptions, and biases of the people who create and use them.

As a leader, I will create a culture where teams recognize their own biases, challenge assumptions, and use AI as a tool to support evidence-based engineering decisions.

HUMAN BIASES CAN SHAPE AI OUTCOMES

Bias can enter at any stage—from data to decisions.



SELECTION BIAS

Using limited or familiar data leads to incomplete models.



CONFIRMATION BIAS

Favoring information that supports what we already believe.



AUTOMATION BIAS

Trusting AI recommendations without enough questioning.



GROUPTHINK & OVERCONFIDENCE

Wanting to agree or being too sure can hide risks.



Awareness is the first step.
Leadership puts safeguards in place.

5 LEADERSHIP STRATEGIES TO NAVIGATE HUMAN BIAS

1



Use Representative Data

I will champion diverse, high-quality data that represents different reservoirs, geologies, operating conditions, and equipment types. Better data leads to fairer, more reliable AI predictions.

In Oil & Gas:

A model trained on one type of reservoir may fail in another. Diverse data improves accuracy across the field.



2



Build Multidisciplinary Teams

I will bring together engineers, geologists, data scientists, environmental experts, and field personnel to challenge assumptions and see blind spots.

In Oil & Gas:

Different perspectives lead to stronger models and better operational decisions.



3



Monitor and Audit Continuously

I will treat AI systems as living solutions. Regular validation and bias audits help us catch performance drift, detect bias, and keep AI trustworthy as conditions change.

In Oil & Gas:

Reservoirs and equipment change over time. Continuous monitoring keeps models reliable.



4



Use Explainable AI

I will promote transparency so engineers understand how AI reaches its conclusions. This reduces automation bias and keeps human expertise in control.

In Oil & Gas:

Understanding why the model made a recommendation helps engineers validate it before taking action.



5



Create a Culture of Learning and Psychological Safety

I will foster an environment where people feel safe to question ideas, challenge AI outputs, and learn from mistakes. This reduces groupthink and drives continuous improvement.

In Oil & Gas:

Open conversations lead to better decisions, safer operations, and a stronger organization.



MY COMMITMENT

I will lead with curiosity, humility, and integrity—ensuring AI in the energy industry is guided by people, purpose, and sound engineering judgment.

“AI doesn’t remove human judgment; it changes where human judgment matters most.”



SOURCE

Chelliah, P. R., Jayasankar, V., Agerstam, M., Sundaravadivazhagan, B., & Cyriac, R. (2023). Explainable artificial intelligence (XAI) for the trust and transparency of the oil and gas systems. In *The Power of Artificial Intelligence for the Next-Generation Oil and Gas Industry*. Wiley.



LEADERSHIP + AWARENESS + COLLABORATION = TRUSTWORTHY AI FOR A SUSTAINABLE ENERGY FUTURE

Pipe Thickness Loss Analysis

Market pipe dataset • 1,000 records • 11 variables

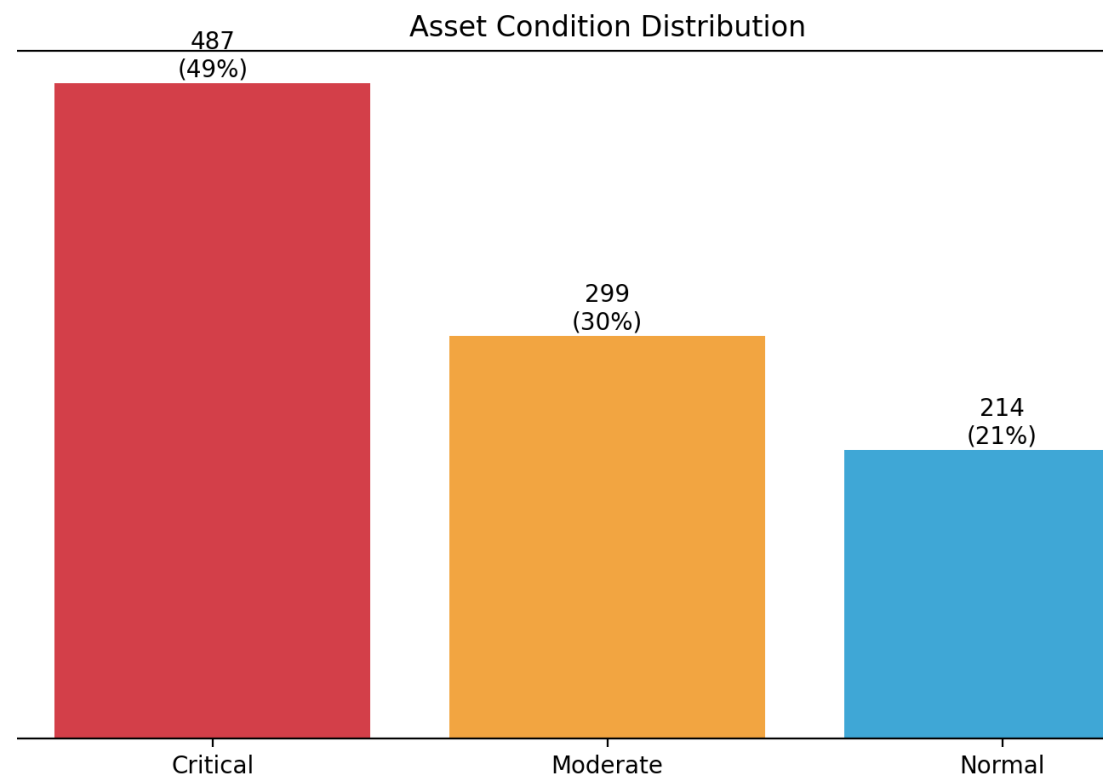
Key headline

Nearly half the pipes are classified Critical, and thickness loss dominates the condition signal.

487 Critical assets

Data quality & profile

The dataset is clean; no missing values or duplicate rows were detected.



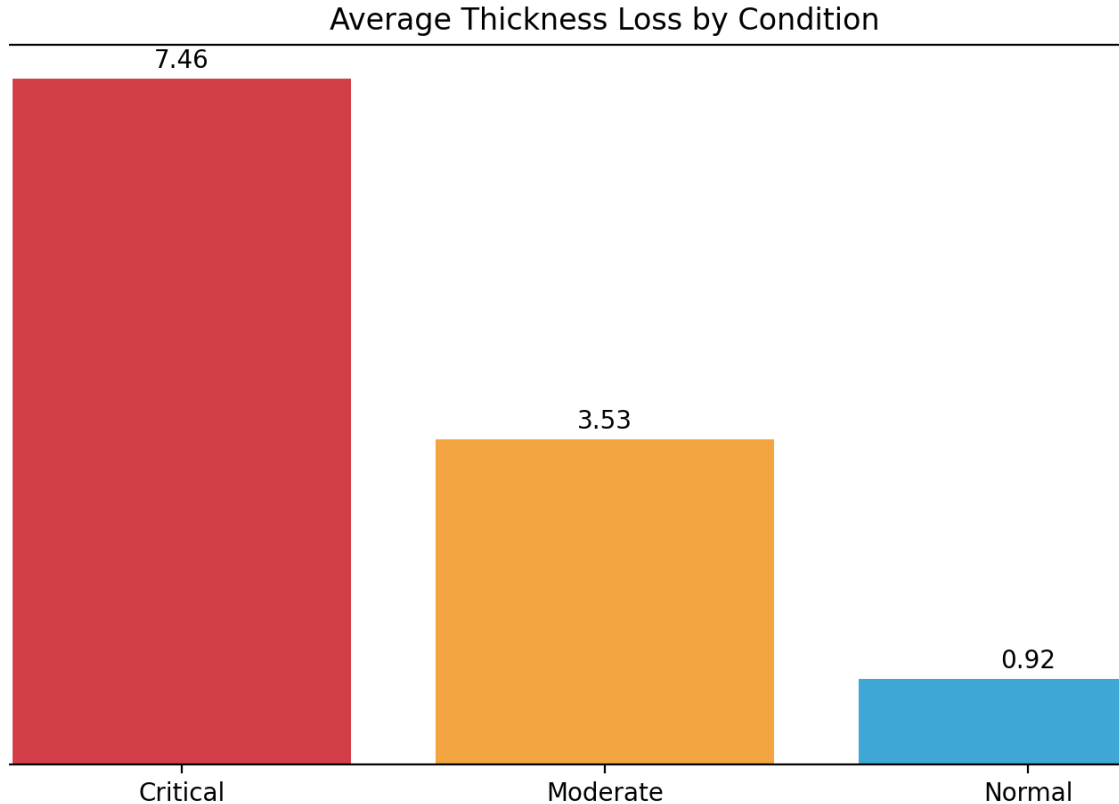
Dataset profile

Records: 1,000
Variables: 11
Materials: 5
Grades: 5
Pipe sizes: 13
Missing values: 0
Duplicate rows: 0

Condition split: 49% Critical, 30% Moderate, 21% Normal

Condition is strongly separated by thickness loss

Average thickness loss rises sharply from Normal to Moderate to Critical.



Interpretation

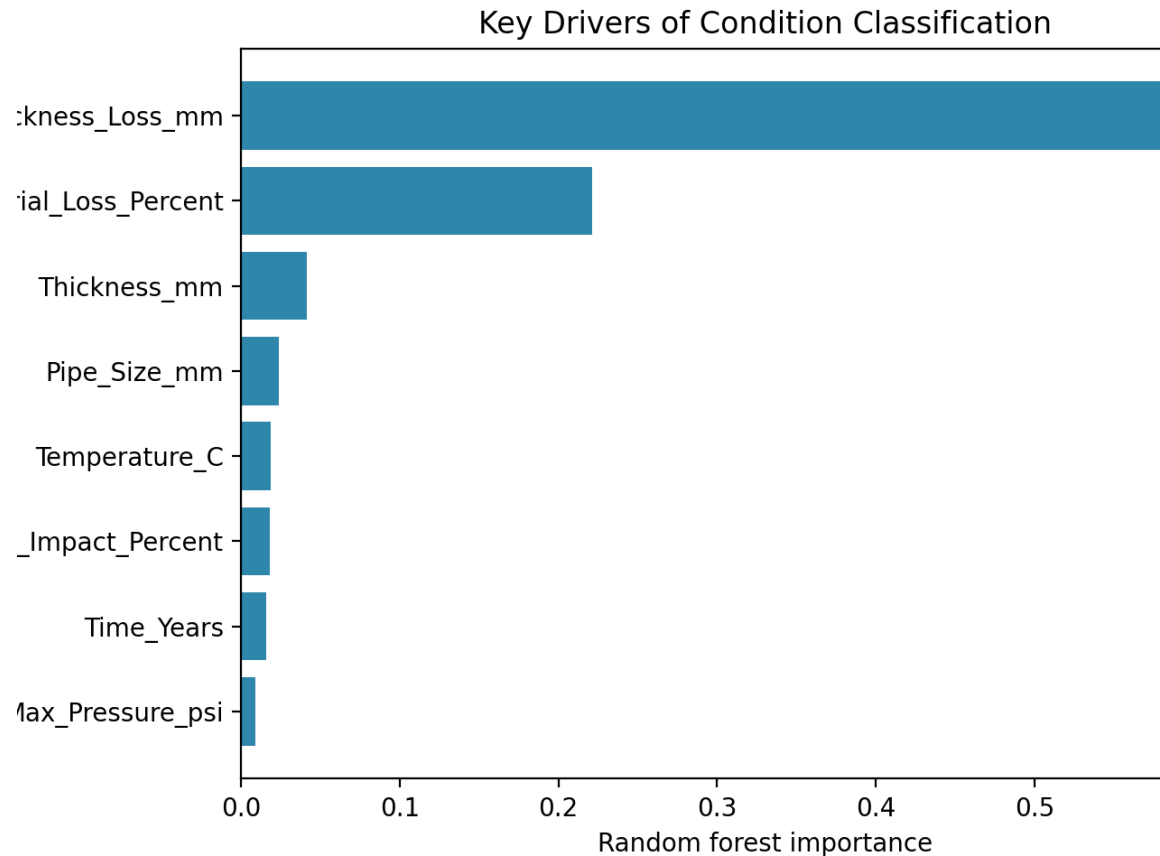
Critical assets average 7.46 mm of thickness loss — more than 8× the Normal average of 0.92 mm.

The practical triage threshold is clear: assets above ~5 mm loss are predominantly Critical in this dataset.

Priority cue: focus inspection capacity on high thickness-loss assets first.

Primary drivers of condition classification

A random-forest classifier confirms that direct loss metrics explain the condition label.



Model readout

5-fold cross-validated accuracy: 99.8%

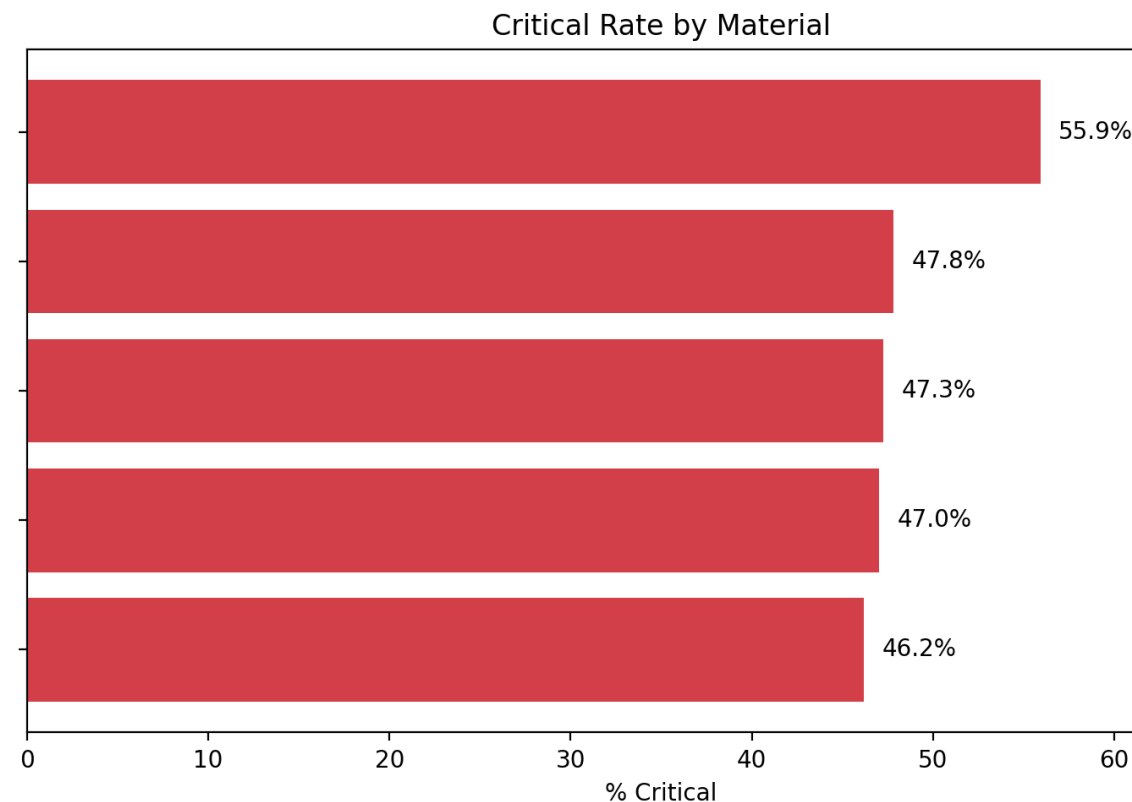
Top drivers:

1. Thickness_Loss_mm
2. Material_Loss_Percent
3. Thickness_mm
4. Pipe_Size_mm

Pressure, temperature, corrosion impact, and age showed weak standalone correlations with thickness loss.

Material and grade risk concentration

PVC has the highest Critical rate; API 5L X65 and X52 rank highest among grades.



Top segments

By material:

PVC: 55.9% Critical

HDPE: 47.8% Critical

Stainless Steel: 47.3% Critical

By grade:

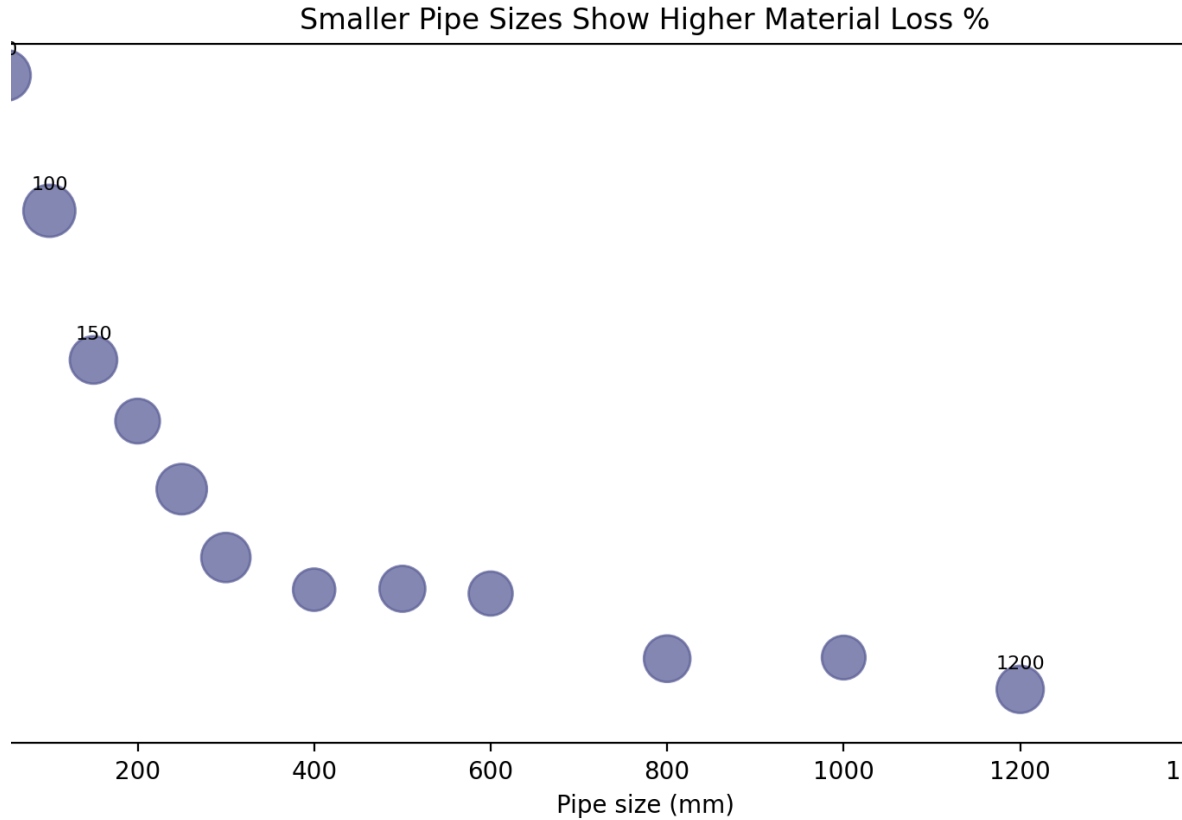
API 5L X65: 53.0% Critical

API 5L X52: 52.9% Critical

Use these segments for risk-based sampling, but avoid over-weighting material alone: direct loss metrics are much stronger indicators.

Pipe size lens: small pipes show higher material loss percentage

Percentage loss is inflated for smaller baseline wall thickness, even when mm loss is similar.



Insight

Average material-loss percentage is highest at 50 mm and 100 mm pipe sizes.

This is because smaller pipes have thinner starting walls; the same millimeter loss translates into a larger percentage loss.

Recommendation: use both absolute mm loss and percentage loss in triage rules.

Recommended action plan — prioritized by risk and consequence

Use condition severity, consequence proxies, and deterioration velocity to sequence work.

1

Tier 1: intervene on high-consequence Critical assets

Inspect the 269 Critical records that also operate at $\geq 1,500$ psi or have diameter ≥ 800 mm. Treat this as the immediate integrity queue.

2

Tier 2: address remaining Critical and near-Critical assets

Schedule the remaining 218 Critical records next, then review 108 Moderate records already at ≥ 4.5 mm loss or $\geq 50\%$ material loss.

3

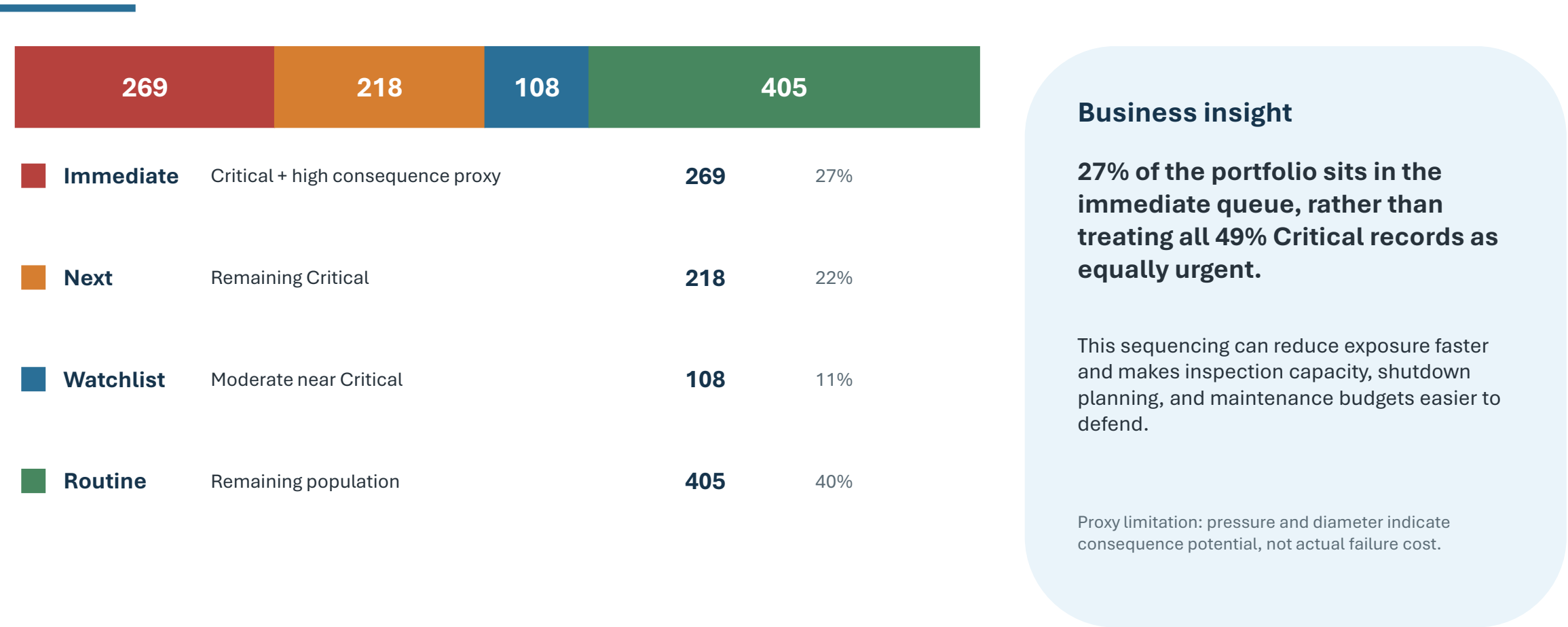
Tier 3: institutionalize risk-based integrity management

Add location, service fluid, coating, inspection dates/method, repair history, shutdown cost, and minimum allowable wall thickness to quantify probability and consequence of failure.

Bottom line: operational risk is concentrated in high-loss assets, not broadly explained by environment variables in this dataset.

A four-tier intervention queue concentrates effort where it matters

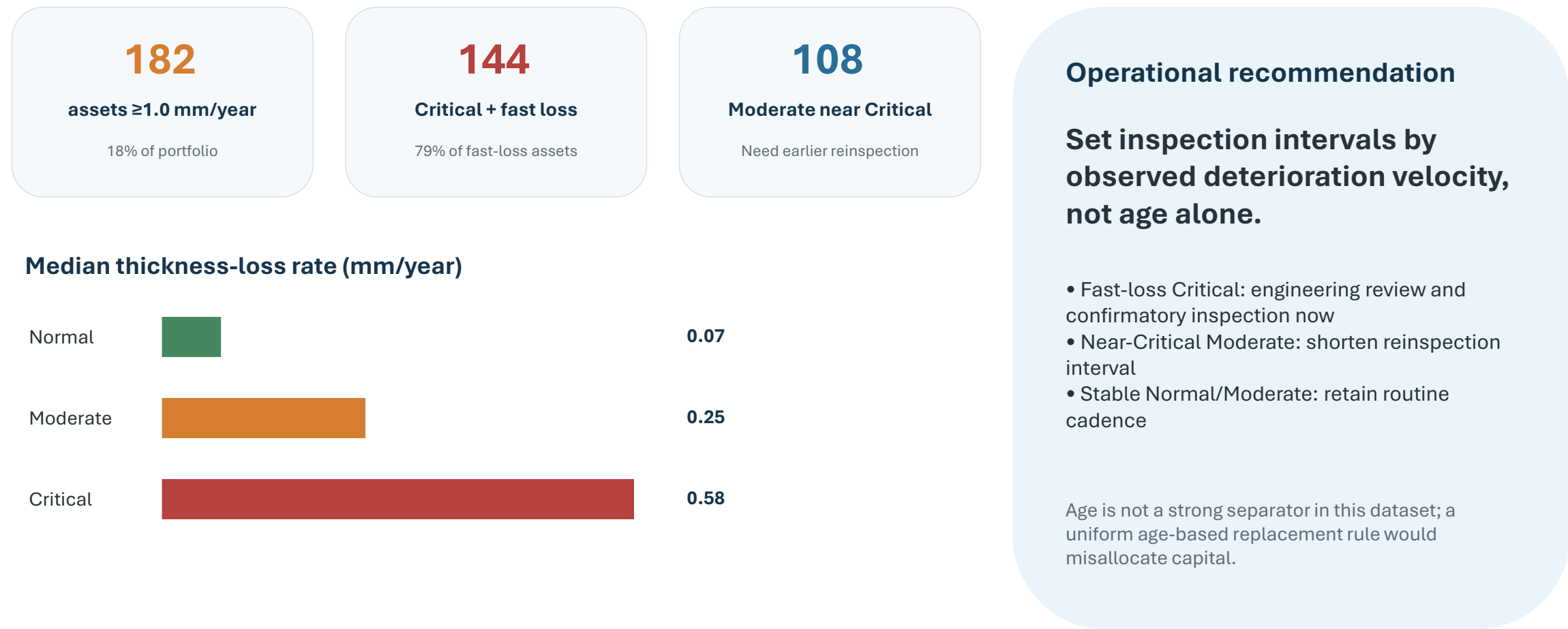
Severity is combined with pressure and diameter as practical consequence proxies.



Decision: reserve emergency/near-term capacity for 269 assets; use planned outages for the next 218 Critical records.

Deterioration velocity identifies assets that may outrun the inspection cycle

Loss per service year is a screening indicator; validate it against actual inspection intervals.



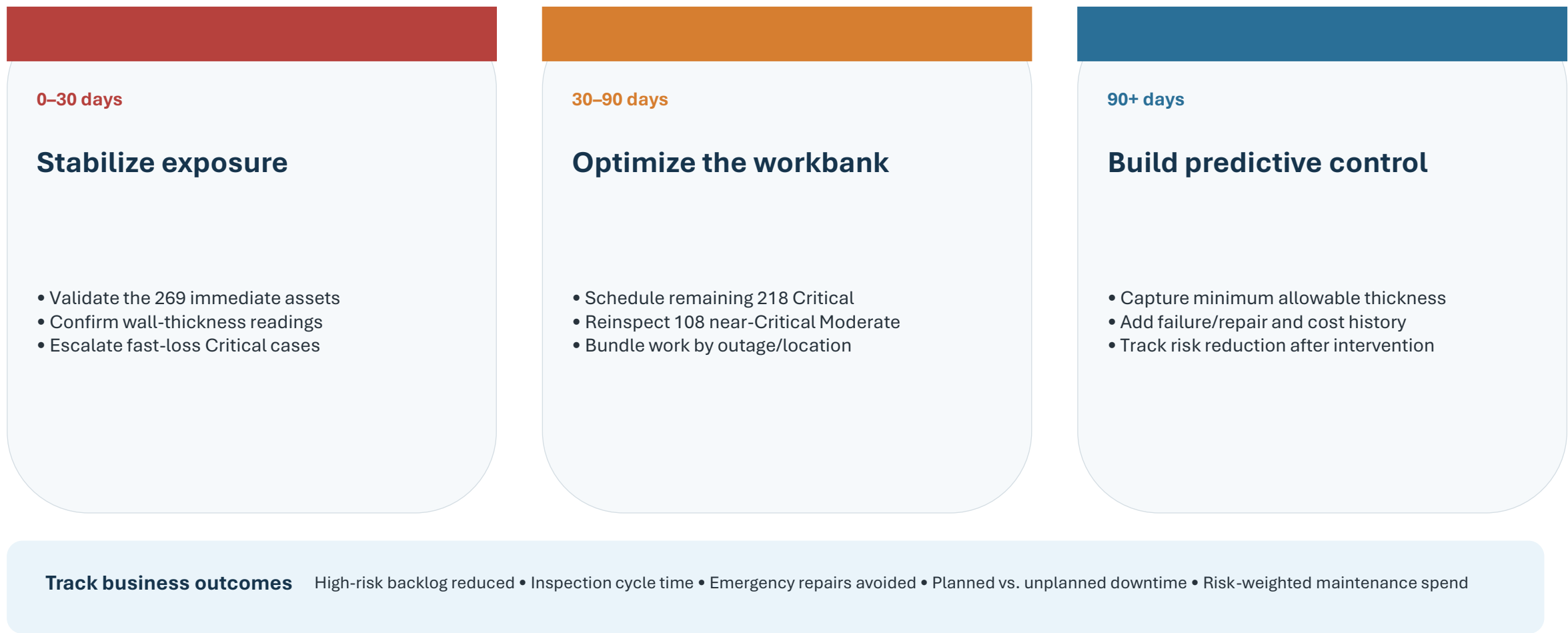
Median thickness-loss rate (mm/year)



Business implication: dynamic intervals can prevent surprise escalation while reducing unnecessary inspections on stable assets.

Convert the analysis into an integrity-management operating model

The dataset supports prioritization; additional fields are needed for defensible failure probability and financial impact.



Do not use this model as a replacement/repair decision by itself; engineering limits and field validation remain required.